Fiscal Year:	FY 2007	**Task Last Updated:**	FY 08/22/2007
PI Name:	van der Woerd, Mark Ph.D.		
Project Title:	The Master Switch for Bone Formation: Structural Studies of RunX2		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	05/01/2006	**End Date:**	09/30/2008
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received (in October 2007) NCE to 9/30/2008 per A. Chu/ARC (8/08)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Lian, Jane (University of Massachusetts Medical School) Snell, Edward (Hauptman-Woodward Medical Research Institute) Ferree, Darren (BAE SYSTEMS Analytical Solutions) Karr, Laura (NASA/MSFC) Stein, Gary (University of Massachusetts Medical School) Montecino, Martin (University of Massachusetts Medical School)		
Grant/Contract No.:	NNA06CB25G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Runx2, which is also called the Core Binding Factor a1 (Cbfa1), is the pivotal molecule for bone formation – the master switch. It regulates osteoblast growth and differentiation at the level of gene transcription, performing these regulatory functions by activation and repression of genes important for bone formation in the osteoblasts. While Runx2 binds to the gene (DNA) with a domain near the N-terminus, the C-terminal region is essential for biological activity. The functional activity of the C-terminus is influenced by a nuclear matrix targeting signal that localizes Runx2 to subnuclear foci for the assembly of transcription factor complexes with a host of interacting co-regulating proteins. We postulate that a significant conformational change occurs within the C-terminal domain of Runx2 upon specific binding of a co-regulatory protein. We hypothesize that the structural changes will differ, dependent on which factor is bound. We are determining both the structure and function of parts of the native and mutant RunX2 molecules, so we will better understand how molecular structure leads to biological function. One of our specific aims is to resolve by X-ray crystallography the three-dimensional structures of the C-terminal domain of wild type and mutant Runx2 molecules, in the absence and presence of regulatory factors. A second specific aim is to experimentally determine the functional activities of Runx2 wild type and point mutants and to identify the mechanisms by which the interactions between Runx2 and other molecules regulate osteoblast differentiation. Significance for NASA: Bone loss occurs at a rate up to 1.5% per month in astronauts while they reside in a reduced gravity environment. It occurs due to an imbalance between bone formation (osteoblast activity on the cellular level) and bone resorption (osteoclast activity). Research on bone loss in long-term reduced gravity missions conducted on astronauts has shown that bone formation is reduced and resorption is unaltered or increased. Our research will help to explain how the activity of the osteoblasts is regulated at the molecular level. Greater understanding of osteoblast activity will lead to future opportunities to better regulate bone formation and therefore reduce bone loss in space and on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Osteoporosis is a recognized disease that affects millions of people, particularly the elderly. It is responsible for hundreds of thousands of fractures annually in the United States alone. Osteoporosis is caused by an imbalance between bone formation and resorption. Each process may be used to restore the balance, for example by anti-resorptive agents or by stimulation of bone formation. The results from our research will contribute to a better understanding of the processes that occur during bone formation. Increasing bone formation is one of the avenues to restore a healthy balance and avoid osteoporosis.
Task Progress:	Our principle findings are as follows: 1) YAP, which is known to bind Runx2 in the C-terminal region, has both positive and negative effects on expression of osteoblast genes. YAP down-regulates Runx2 promoter activity and can repress Runx2 activity on Runx2 target genes, such as osteocalcin that is a significant component of bone matrix. The disruption of the Runx2-YAP interactions increases the availability of Runx2 protein in the cell for trans-activation of other genes. 2) However, YAP also has the ability to promote transcription of osteoblast genes independent of Runx2. YAP appears to affect transcriptional activity of some bone gene promoters independent of Runx2 through a protein-protein interaction with possibly another transcription factor on the osteocalcin gene. 3) In the course of our studies, we and others have identified several WW domain proteins, such as WWOX and WWP1, which interact with Runx2 and affect its activity in osteoblasts. 4) A major discovery during this program was that Runx1 and YAP interact in the bone resorbing osteoclast cells. The proteins Runx1 and Runx2 are highly conserved, especially in the PPxY motif, which interacts with the YAP-WW domain. We have found that many genes essential for osteoclast adhesion and activity of the osteoclast in degrading bone contain Runx regulatory elements. Thus disruption of Runx1 activity could reduce osteolytic bone disease. 5) We are now able to mass-produce large quantities of purified (partial) YAP and purified (partial) Runx2, using genetic manipulation of E. coli. We are using these proteins for biophysical characterization and for structural studies with the intention to explain the function of YAP and Runx protein, specifically by studying what changes occur upon binding of YAP to Runx2.
Bibliography Type:	Description: (Last Updated: 08/23/2007)
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Articles in Peer-reviewed Journals	Saltman LH, Javed A, Ribadeneyra J, Hussain S, Young DW, Osdoby P, Amcheslavsky A, van Wijnen AJ, Stein JL, Stein GS, Lian JB, Bar-Shavit Z. "Organization of transcriptional regulatory machinery in osteoclast nuclei: Compartmentalization of Runx1." J Cell Physiol. 2005 Sep;204(3):871-80. PMID: 15828028 , Sep-2005
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